

Numerical Methods

By

Numerical Methods Group

Chapter (4): Numerical Differentiation and Integration

Numerical Integration

Numerical integration is used to approximate the value of a definite integral when:

- The function is difficult or impossible to integrate analytically
- Data is given in tabular form
- The function is known only at discrete points

we present the following methods of integration with illustrative examples:

1. Trapezoidal rule.
2. Simpson's $1/3$ rule.
3. Simpson's $3/8$ rule.
4. Boole's and Weddle's rules.

Trapezoidal Rule:

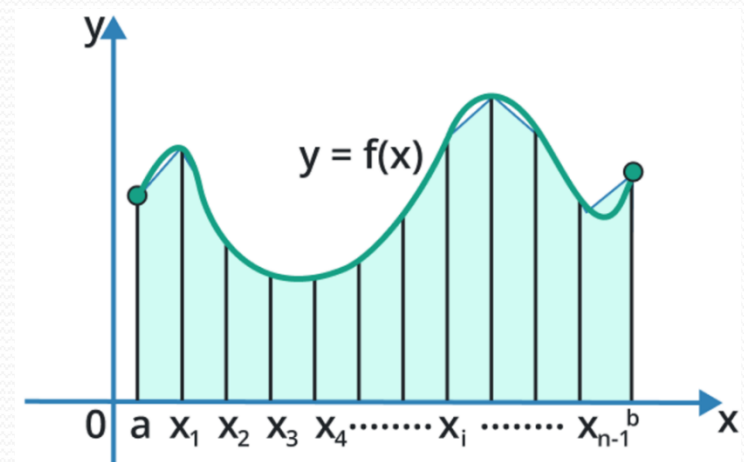
The Trapezoidal Rule approximates the definite integral by dividing the area under the curve into trapezoids instead of rectangles

$$\int_a^b f(x) dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n]$$

Where:

$$h = \frac{b - a}{n}$$

$$y_i = f(x_i)$$



Example :

Evaluate the integral $\int_0^{1.2} e^x dx$, taking six intervals by using trapezoidal rule up to three significant figures.

Solution :

$$a = 0, b = 1.2, n = 6$$
$$h = \frac{b - a}{n} = \frac{1.2 - 0}{6} = 0.2$$

x	0	0.2	0.4	0.6	0.8	1	1.2
y	0	1.221	1.492	1.822	2.226	2.718	3.320

The trapezoidal rule can be written as:

$$\int_a^b f(x) dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n]$$

$$\int_0^{1.2} e^x dx = \frac{0.2}{2} [0 + 2(1.221 + 1.492 + 1.822 + 2.226 + 2.718) + 3.320] = 2.328$$

The exact value is:

$$\int_0^{1.2} e^x dx = e^x \Big|_0^{1.2} = e^{1.2} - e^0 = 2.320$$

Example :

Evaluate the integral $\int_0^{12} \frac{1}{1+x^2} dx$, using trapezoidal rule taking $n = 6$, correct to give significant figures.

Solution :

$$a = 0, b = 12, n = 6$$

$$h = \frac{b - a}{n} = \frac{12 - 0}{6} = 2$$

x	0	2	4	6	8	10	12
y	1	$\frac{1}{5}$	$\frac{1}{17}$	$\frac{1}{37}$	$\frac{1}{65}$	$\frac{1}{101}$	$\frac{1}{145}$

The trapezoidal rule can be written as:

$$\int_a^b f(x) dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n]$$

$$\int_0^{12} \frac{1}{1+x^2} dx = \frac{2}{2} \left[1 + 2 \left(\frac{1}{5} + \frac{1}{17} + \frac{1}{37} + \frac{1}{65} + \frac{1}{101} \right) + \frac{1}{145} \right] = 1.62916$$

The exact value is:

$$\int_0^{12} \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^{12} = \tan^{-1} 12 - \tan^{-1} 0 = 1.48766$$

Example :

Evaluate the integral $\int_2^6 \log x \, dx$, using trapezoidal rule taking $n = 8$, correct to five decimal places.

Solution :

Try by yourself

Simpson's 1/3 rule:

Simpson's 1/3 Rule is a highly accurate method that approximates the integral using parabolic curves instead of straight lines.

$$\int_a^b f(x)dx = \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5 + \cdots) + 2(y_2 + y_4 + y_6 + \cdots) + y_n]$$

Conditions:

- Number of intervals n must be even
- Data must be equally spaced
- $h = \frac{b-a}{n}$, $y_i = f(x_i)$

Example :

Evaluate the integral $\int_0^{1.2} e^x dx$, taking $n = 6$ using Simpson's $1/3$ rule.

Solution :

$$a = 0, b = 1.2, n = 6$$

$$h = \frac{b - a}{n} = \frac{1.2 - 0}{6} = 0.2$$

x	0	0.2	0.4	0.6	0.8	1	1.2
y	0	1.221	1.492	1.822	2.226	2.718	3.320

The Simpson's rule is:

$$\int_a^b f(x)dx = \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5 + \cdots) + 2(y_2 + y_4 + y_6 + \cdots) + y_n]$$

$$\begin{aligned} \int_0^{1.2} e^x dx &= \frac{0.2}{3} [0 + 4(1.221 + 1.822 + 2.718) + 2(1.492 + 2.226) + 3.320] \\ &= 2.32014 \end{aligned}$$

The exact value is:

$$\int_0^{1.2} e^x dx = e^x \Big|_0^{1.2} = e^{1.2} - e^0 = 2.320$$

Example :

Evaluate the integral $\int_0^{12} \frac{1}{1+x^2} dx$, taking $n = 6$ using Simpson's 1/3 rule.

Solution :

$$a = 0, b = 12, n = 6$$

$$h = \frac{b - a}{n} = \frac{12 - 0}{6} = 2$$

x	0	2	4	6	8	10	12
y	1	$\frac{1}{5}$	$\frac{1}{17}$	$\frac{1}{37}$	$\frac{1}{65}$	$\frac{1}{101}$	$\frac{1}{145}$

The Simpson's rule is:

$$\int_a^b f(x)dx = \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5 + \cdots) + 2(y_2 + y_4 + y_6 + \cdots) + y_n]$$

$$\int_0^{12} \frac{1}{1+x^2} dx = \frac{2}{3} \left[1 + 4 \left(\frac{1}{5} + \frac{1}{37} + \frac{1}{101} \right) + 2 \left(\frac{1}{17} + \frac{1}{65} \right) + \frac{1}{145} \right] = 1.40201$$

The exact value is:

$$\int_0^{12} \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^{12} = \tan^{-1} 12 - \tan^{-1} 0 = 1.48766$$

Example :

Evaluate the integral $\int_2^6 \log x \, dx$, taking $n = 6$ using Simpson's $1/3$ rule.

Solution :

Try by yourself

Simpson's 3/8 rule:

Simpson's 3/8 Rule is a numerical method that approximates the integral using cubic (third-degree) polynomials. It is more accurate than the trapezoidal rule and useful when Simpson's 1/3 condition is not satisfied.

$$\int_a^b f(x)dx = \frac{3h}{8} [y_0 + 3(y_1 + y_2 + y_4 + y_5 + \cdots) + 2(y_3 + y_6 + \cdots) + y_n]$$

Conditions:

- Number of intervals n must be a multiple of 3
- Data must be equally spaced
- $h = \frac{b-a}{n}$, $y_i = f(x_i)$

Example :

Evaluate the integral $\int_0^{1.2} e^x dx$, taking $n = 6$ using Simpson's 3/8 rule.

Solution :

$$a = 0, b = 1.2, n = 6$$

$$h = \frac{b - a}{n} = \frac{1.2 - 0}{6} = 0.2$$

x	0	0.2	0.4	0.6	0.8	1	1.2
y	0	1.221	1.492	1.822	2.226	2.718	3.320

The Simpson's rule is:

$$\int_a^b f(x)dx = \frac{3h}{8} [y_0 + 3(y_1 + y_2 + y_4 + y_5 + \cdots) + 2(y_3 + y_6 + \cdots) + y_n]$$

$$\begin{aligned} \int_0^{1.2} e^x dx &= \frac{3(0.2)}{8} [0 + 3(1.221 + 1.492 + 2.226 + 2.718) + 2(1.822) + 3.320] \\ &= 2.320 \end{aligned}$$

The exact value is:

$$\int_0^{1.2} e^x dx = e^x \Big|_0^{1.2} = e^{1.2} - e^0 = 2.320$$

Example :

Evaluate the integral $\int_0^{12} \frac{1}{1+x^2} dx$, taking $n = 6$ using Simpson's 3/8 rule

Solution :

$$a = 0, b = 12, n = 6$$

$$h = \frac{b - a}{n} = \frac{12 - 0}{6} = 2$$

x	0	2	4	6	8	10	12
y	1	$\frac{1}{5}$	$\frac{1}{17}$	$\frac{1}{37}$	$\frac{1}{65}$	$\frac{1}{101}$	$\frac{1}{145}$

The Simpson's rule is:

$$\int_a^b f(x)dx = \frac{3h}{8} [y_0 + 3(y_1 + y_2 + y_4 + y_5 + \cdots) + 2(y_3 + y_6 + \cdots) + y_n]$$

$$\int_0^{12} \frac{1}{1+x^2} dx = \frac{3(2)}{8} \left[1 + 3 \left(\frac{1}{5} + \frac{1}{17} + \frac{1}{65} + \frac{1}{101} \right) + 2 \left(\frac{1}{37} \right) + \frac{1}{145} \right] = 1.43495$$

The exact value is:

$$\int_0^{12} \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^{12} = \tan^{-1} 12 - \tan^{-1} 0 = 1.48766$$

Example :

Evaluate the integral $\int_2^6 \log x \, dx$, taking $n = 6$ using Simpson's 3/8 rule.

Solution :

Try by yourself

Boole's and Weddle's rules:

Boole's Rule and Weddle's Rule are numerical integration techniques (Newton–Cotes formulas) used to approximate definite integrals when the function is known at equally spaced points.

Boole's Rule:

Boole's Rule is a 5-point closed Newton–Cotes formula and taking the curve $y = f(x)$ through (x_n, y_n) , $n = 0, 1, 2, 3, 4$ as a polynomial of degree 4

$$\int_a^b f(x)dx = \frac{2h}{25} [7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4]$$

Conditions:

- Number of intervals 4
- Data must be **equally spaced**
- $h = \frac{b-a}{4}$, $y_i = f(x_i)$

Weddle's Rule:

Weddle's Rule is a 7-point closed Newton–Cotes formula and taking the curve $y = f(x)$ through (x_n, y_n) , $n = 0, 1, 2, 3, 4, 5, 6$ as a polynomial of degree 6

$$\int_a^b f(x)dx = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6]$$

Conditions:

- Number of intervals 6
- Data must be **equally spaced**
- $h = \frac{b-a}{6}$, $y_i = f(x_i)$

Example :

Evaluate the integral $\int_0^{1.2} e^x dx$, using Boole's rule using exactly five functional evaluations and correct to five significant figures.

Solution :

$$a = 0, b = 1.2, n = 4$$
$$h = \frac{b - a}{n} = \frac{1.2 - 0}{4} = 0.3$$

x	0	0.3	0.6	0.9	1.2
y	1	1.350	1.822	2.460	3.320

The Boole's rule is:

$$\int_a^b f(x)dx = \frac{2h}{25} [7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4]$$

$$\begin{aligned} \int_0^{1.2} e^x dx &= \frac{2(0.3)}{25} [7 + 32(1.350) + 12(1.822) + 32(2.460) + 7(3.320)] \\ &= 2.3195 \end{aligned}$$

The exact value is:

$$\int_0^{1.2} e^x dx = e^x \Big|_0^{1.2} = e^{1.2} - e^0 = 2.320$$

Example :

Evaluate the integral $\int_0^{12} \frac{1}{1+x^2} dx$, using Boole's rule using exactly five functional evaluations and correct to five significant figures.

Solution :

$$a = 0, b = 12, n = 4$$

$$h = \frac{b - a}{n} = \frac{12 - 0}{4} = 3$$

x	0	3	6	9	12
y	1	$\frac{1}{10}$	$\frac{1}{37}$	$\frac{1}{82}$	$\frac{1}{145}$

The Boole's rule is:

$$\int_a^b f(x)dx = \frac{2h}{25} [7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4]$$

$$\int_0^{12} \frac{1}{1+x^2} dx = \frac{2(3)}{25} \left[1 + 32 \left(\frac{1}{10} \right) + 12 \left(\frac{1}{37} \right) + 32 \left(\frac{1}{82} \right) + 7 \left(\frac{1}{145} \right) \right] = 1.4617$$

The exact value is:

$$\int_0^{12} \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^{12} = \tan^{-1} 12 - \tan^{-1} 0 = 1.48766$$

Example :

Evaluate the integral $\int_0^{1.2} e^x dx$, using Weddle's rule and taking $n = 6$, correct to five significant figures.

Solution :

$$a = 0, b = 1.2, n = 6$$
$$h = \frac{b - a}{n} = \frac{1.2 - 0}{6} = 0.2$$

x	0	0.2	0.4	0.6	0.8	1	1.2
y	1	1.221	1.491	1.822	2.226	2.718	3.320

The Weddle's rule is:

$$\int_a^b f(x)dx = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6]$$

$$\begin{aligned} \int_0^{1.2} e^x dx &= \frac{3(0.2)}{10} [1 + 5(1.221) + 1.491 + 6(1.822) + 2.226 + 5(2.718) + 3.320] \\ &= 2.32011 \end{aligned}$$

The exact value is:

$$\int_0^{1.2} e^x dx = e^x \Big|_0^{1.2} = e^{1.2} - e^0 = 2.320$$

Example :

Evaluate the integral $\int_0^{12} \frac{1}{1+x^2} dx$, using Weddle's rule and taking $n = 6$, correct to five significant figures.

Solution :

$$a = 0, b = 12, n = 6$$

$$h = \frac{b - a}{n} = \frac{12 - 0}{6} = 2$$

x	0	2	4	6	8	10	12
y	1	$\frac{1}{5}$	$\frac{1}{17}$	$\frac{1}{37}$	$\frac{1}{65}$	$\frac{1}{101}$	$\frac{1}{145}$

The Weddle's rule is:

$$\int_a^b f(x)dx = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6]$$

$$\begin{aligned} \int_0^{12} \frac{1}{1+x^2} dx &= \frac{3(2)}{10} \left[1 + 5 \left(\frac{1}{5} \right) + \left(\frac{1}{17} \right) + 6 \left(\frac{1}{37} \right) + \left(\frac{1}{65} \right) + 5 \left(\frac{1}{101} \right) + \left(\frac{1}{145} \right) \right] \\ &= 1.37567 \end{aligned}$$

The exact value is:

$$\int_0^{12} \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^{12} = \tan^{-1} 12 - \tan^{-1} 0 = 1.48766$$

Example :

Evaluate the integral $\int_2^6 \log x \, dx$, using:

- 1- Boole's rule
- 2- Weddle's rule

Solution :

Try by yourself

**THANK YOU
FOR YOUR
ATTENTION**

